BlurGuard - Class Diagram

User Management

User
- userId : int
- email : String
+ logout()
+ login()

History and Reporting

Presenter
+ activatePanicProtection()
+ startSession()
+ pauseSession()
+ stopSession()

Administrator
+ managePolicies()
+ viewLogs()
+ manageUsers()

Detection
- detectionId : int
- dataType : String
- confidence : float
- location : String

ScreeningSession
- sessionId : int
- status : String
+ start()
+ pause()
+ resume()
+ stop()

DetectionHistory
- recordedAt : DateTime
+ viewHistory()
+ saveDetection()

Report
- reportId : int
- generatedAt : DateTime
+ generateReport()

AuditLog
- logId : int
- action : String
- timestamp : DateTime
+ recordAction()

Protection and Output

RiskAssessment
- riskLevel: String
- riskScore: float
+ calculateRisk()

PanicProtection
+ blockOutput()
+ activate()
+ deactivate()

ProtectedOutput
+ outputStatus : String
+ generateOutput()
+ updatePreview()

Screen Capture

ScreenSource
- sourceName : String
- sourceType : String
+ selectSource()

Detection and Risk

Frame
- frameId: int
- timeStamp : DateTime

Policies and Configuration

ProtectionPolicy
- policyId : int
- name : String
- maskingStyle : String
+ updatePolicy()

SensitivePattern
- patternId : int
- patternType : String
+ addPattern()
+ removePattern()

DetectionModel
- modelName : String
- version : String
+ loadModel()

ApplicationRule
- applicationName : String
- protectionEnabled : Boolean
+ updateRule()

DetectionEngine
+ detectSensitiveData()
+ scanFrame()

MaskingEngine
- maskingStyle : String
+ redactData()
+ applyBlur()
+ adjustMask()

SharingApplication
- applicationName : String
+ sendProtectedOutput()

starts
views
records
activates
1
has
selects
reviews
captures
provides data
configures
records activity
manages
identifies regions
guides masking
supports
controls
analyzes
supplies patterns
delivers
identifies
creates
applyBlur()